

Physics engine vs. machine-learning surrogates for PC-SAFT interfacial properties: a timing and accuracy benchmark

Generated benchmark report

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Abstract

We benchmark three families of models that predict interfacial and phase properties of multicomponent mixtures described by the PC-SAFT Helmholtz energy functional: (i) the physics engine itself, classical density functional theory (cDFT) solved with **Fe0s**; (ii) *full data-driven* surrogates based on the TabPFN tabular foundation model (two variants: **without_HP0** and **with_HP0**); and (iii) *residual-correction* models (Gaussian process regression, GPR; stochastic variational GP, SVGP; and symbolic regression, SR) that learn the deviation of cDFT from an analytic baseline. Across all four targets (bubble pressure P_{bubble} , dew pressure P_{dew} , surface tension γ , and interfacial thickness) the surrogates reproduce cDFT essentially exactly ($R^2 \gtrsim 0.9999$ on a held-out test split) while reducing the per-state-point cost from *seconds* (cDFT) to *sub-millisecond* (TabPFN, GPR, SR), a speed-up of 10^3 – $10^4\times$. Accuracy is reported as 5-fold cross-validated (CV) RMSE — the honest generalisation estimate, systematically larger than the i.i.d. test error. We place the timings in the context of the GPU-accelerated cDFT of Stierle et al. (2024) and the **Fe0s** reference of Rehner et al. (2023), and show by direct re-timing on the same node that the multicomponent interface solve — *not* the phase-equilibrium flash — is what makes our per-point cDFT cost exceed the simple-fluid reference.

1 Introduction and scope

Classical density functional theory (cDFT) based on the PC-SAFT equation of state is a quantitatively accurate route to interfacial properties such as surface tension, but it is computationally expensive: each state point requires the iterative solution of the Euler–Lagrange equation for the density profile, dominated by the convolution integrals of the weighted-density functional [1, 2]. This cost is prohibitive for the large-scale screening and inverse-design tasks that motivate the field [2].

A machine-learning (ML) surrogate trained on cDFT data can, in principle, collapse this cost to a single forward pass. We quantify this trade-off for two surrogate strategies:

- **Full data-driven models** that map the thermodynamic state $(T, P, \mathbf{z})$ directly to the target property, using the TabPFN tabular foundation model in two variants (**without_HP0** and a hyperparameter-optimised **with_HP0**).
- **Residual-correction models** that predict only the *residual* between cDFT and a cheap analytic baseline (γ_{wsd}), which the physics already captures most of. We use GPR, SVGP, and symbolic regression (SR).

Accuracy metric. Unless stated otherwise, the headline accuracy figure is the **5-fold cross-validated RMSE** (mean $\pm$ standard deviation over folds). CV is reported because the held-out i.i.d. test split is optimistic for this data — the folds force the model to generalise across state regions, so the CV RMSE is the number that matches real deployment error (Sec. 5).

Hardware and environment. Three distinct machines are used; cross-machine timings are therefore order-of-magnitude, not same-rig (Table 1). The cDFT γ maps and the per-stage re-timing (Sec. 5, Table 9) ran with **Fe0s** (CPU, BLAS-threaded) on the TU Delft **c109 highmem** node (2 $\times$ AMD EPYC 9754, 256 cores, $\approx$ 1.5 TB RAM). TabPFN training/inference was timed on an NVIDIA H100 GPU (Snellius). The residual models (GPR/SVGP/SR) were timed in the **py_A6** environment (PyTorch 2.10 CPU, GPyTorch 1.15.2, scikit-learn 1.6.1) on a local Intel Xeon Silver 4214R @ 2.40 GHz (2 sockets $\times$ 12 cores = 24 threads). All learned models share the training split $n_{\text{train}} = 13\,552$, $n_{\text{test}} = 2\,904$ (per-property validation split of comparable size); CV is 5-fold on the pooled data.

Table 1: Compute hardware used for each part of the benchmark.

Machine	Hardware	Used for
c109 (TU Delft)	2 $\times$ AMD EPYC 9754, 256 cores, $\approx$ 1.5 TB RAM (CPU-only)	cDFT maps + per-stage re-timing
Snellius	NVIDIA H100 GPU	TabPFN training / inference
Local workstation	Intel Xeon Silver 4214R @ 2.40 GHz, 24 threads	GPR / SVGP / SR residual-model

2 Full data-driven surrogates: TabPFN across all targets

2.1 Accuracy

Table 2 reports the test/validation accuracy and the 5-fold CV RMSE of the two TabPFN variants for all four targets. On the i.i.d. test split both variants reproduce cDFT with $R^2 \geq 0.9998$; **with_HP0** is the more accurate everywhere and is the deployed model. Note the CV RMSE is markedly larger than the test RMSE (e.g. γ **with_HP0**: CV 0.040 vs test 0.006 mN m $^{-1}$) — the realistic generalisation cost.

2.2 Cost: training, inference, memory, model size

Table 3 summarises the micro-benchmark of the two variants (mean over the four targets). The trade-off is clear:

- **without_HP0**: cheap to train ($\sim$ 1.5 s) *and* to serve ($\sim$ 0.7–1.7 ms/sample); lightest GPU footprint.
- **with_HP0**: expensive training ($\sim$ 17 min of HPO search) but fast, light-weight serving; the deployed γ model.

3 cDFT vs. the full data-driven surrogate on a γ map

Table 4 compares the cost of generating a γ map ($\sim$ 10 000 $(T, P, \mathbf{z})$ points per feed) with cDFT against the deployed **with_HP0** TabPFN model predicting the *identical* points. The cDFT map takes 2–18 h; the surrogate takes 3.5–6.9 s, a speed-up of 10^3 – $10^4\times$ at no meaningful loss of accuracy ($R^2 \approx 0.9999$, RMSE $\approx$ 0.03–0.05 mN m $^{-1}$).

4 Residual-correction models for γ

Instead of learning γ directly, the residual models learn the deviation of cDFT from the analytic baseline γ_{wsd} , so the reconstructed prediction is $\gamma_{\text{pred}} = \gamma_{\text{base}} + \hat{r}$. This keeps the physics in the loop and leaves only a small, smooth residual to be learned. Table 5 reports their accuracy. GPR (5000-point training subset) and SVGP (2000 inducing points) target the residual directly

Table 2: Accuracy of the TabPFN variants on the held-out test and validation sets, and 5-fold cross-validated RMSE. RMSE units: pressures in bar, γ in mN m^{-1} , interfacial thickness in its native unit. RMSE_{CV} is the mean $\pm$ standard deviation over the 5 folds (same units). Bold marks the lower CV RMSE in each target group.

Target	Variant	R^2_{test}	R^2_{val}	$\text{RMSE}_{\text{test}}$	RMSE_{val}	RMSE_{CV} (mean $\pm$ std)
P_{bubble}	without_HPO	0.999930	0.999926	0.18249	0.18249	2.732 ± 0.587
	with_HPO	0.999997	0.999997	0.03582	0.03767	1.313 ± 0.302
P_{dew}	without_HPO	0.999947	0.999941	0.13515	0.13812	0.298 ± 0.077
	with_HPO	1.000000	1.000000	0.00214	0.00198	0.201 ± 0.184
γ	without_HPO	0.999956	0.999961	0.04062	0.03828	0.1630 ± 0.0172
	with_HPO	0.999999	0.999999	0.00603	0.00600	0.0395 ± 0.0094
thickness	without_HPO	0.999850	0.999823	0.00843	0.00896	0.0272 ± 0.0143
	with_HPO	0.999896	0.999909	0.00702	0.00642	0.0277 ± 0.0081

Table 3: TabPFN variant micro-benchmark (mean over the four targets; $n_{\text{train}} = 13\,552$, $n_{\text{test}} = 2\,904$; H100 GPU).

Metric	without_HPO	with_HPO
Training wall time [s]	~ 1.5	~ 1008 (≈ 17 min)
Inference [ms/sample]	0.7–1.7	0.5–5.4
GPU memory [MB]	50–86	~ 4900
Model size [MB]	87.9	~ 51

(RMSE in mN m^{-1}); SR targets a dimensionless $\varepsilon_{\text{base}}$ and we quote its reconstructed γ_{cDFT} -scale RMSE. The CV RMSE columns are sourced from the cross-validated runs of each model, so test, validation and CV in a row refer to one and the same fitted model.

5 Cross-validated RMSE for γ

Table 6 collects the 5-fold CV RMSE for γ across every learned model on a common mN m^{-1} scale. Reading it requires one caveat: TabPFN predicts γ *directly*, whereas GPR and SVGP predict only the small *residual* $\gamma_{\text{cDFT}} - \gamma_{\text{wsd}}$, so their RMSE is on a quantity of much smaller magnitude — the residual numbers are therefore not directly comparable to the direct- γ TabPFN numbers, even though both are in mN m^{-1} . Among the full data-driven models, **with_HPO** TabPFN is the best at 0.040 mN m^{-1} . The residual SVGP has the smallest absolute CV RMSE (0.023 mN m^{-1}), as expected for the easier residual target.

Crucially, the CV RMSE of the deployed **with_HPO** model (0.040 mN m^{-1}) *agrees with* its deployment RMSE when rolled out over the full cDFT γ map (0.03 – 0.05 mN m^{-1} , Table 4), whereas the i.i.d. test RMSE (0.006 mN m^{-1}) was a factor ~ 6 optimistic. This is exactly why we headline the CV figure.

6 Per-state-point timing: the unifying comparison

Table 7 is the headline comparison: the wall time to evaluate γ at one state point, for the physics engine and every surrogate. “Isolated” is a single-sample call (includes call overhead); “batched” is the amortised per-point cost when predicting the whole map at once (the realistic deployment cost).

Table 4: cDFT (PC-SAFT, CPU/BLAS) vs. `with_HP0` TabPFN (H100, inference only) on the same per-feed γ map. Speed-up grows with feed because cDFT cost varies strongly while ML inference is nearly constant.

Feed	Points	cDFT map [s]	ML map [s]	Speed-up	R^2	RMSE [mN m ⁻¹]
0	9 987	7 481.1 (2.1 h)	6.89	1 086×	0.99997	0.035
1	9 764	19 253.2 (5.3 h)	5.73	3 359×	0.99998	0.026
2	10 120	18 286.8 (5.1 h)	3.56	5 136×	0.99995	0.046
3	9 910	65 151.9 (18.1 h)	4.34	15 026×	0.99997	0.032
4	10 050	63 255.5 (17.6 h)	3.54	17 874×	0.99996	0.038

Table 5: Residual-correction models for γ . RMSE_{CV} is the 5-fold cross-validated mean $\pm$ std. GPR/SVGP RMSE is on the residual target (mN m⁻¹); SR RMSE is on the reconstructed γ_{cDFT} scale (mN m⁻¹). R_{CV}^2 is on the same target as the RMSE.

Model	Target	$\text{RMSE}_{\text{test}}$	RMSE_{val}	RMSE_{CV} (mean $\pm$ std)	R_{CV}^2
GPR (RBF+White, 5000)	residual	0.01584	0.01536	0.0379 ± 0.0262	0.9938
SVGP (Matérn-5/2 ARD)	residual	0.03684	0.02792	0.0233 ± 0.0030	0.9983
SR (closed-form eq.)	γ_{cDFT}	0.14181	0.14235	0.2475 ± 0.0110	0.9983

6.1 Training, inference, memory and model size

Table 8 compares the three model families on the same resource axes as a conventional surrogate benchmark: training cost, per-sample inference cost, model size, and peak memory. The physics engine (PC-SAFT + cDFT) has no training phase and no stored model — its “inference” is the full per-point solve — whereas the learned models pay a one-off training cost and then serve predictions at $\sim 1\text{--}5$ ms/sample (isolated) or far less when batched.

7 Comparison with the literature

Our cDFT cost should be read against two reference implementations.

Where the per-point seconds actually go. To make our per-point cost comparable to literature figures (which time a single simple-fluid interface), we re-ran the V5 contour-map sweep for the five study feeds on the same c109 node, instrumented to separate the *phase-equilibria* stage (critical point + bubble/dew envelope, once per feed) from the *cDFT interfacial sweep* (TP-flash + planar-interface solve at every (T, P) point), at the default grid ($n_{\text{grid}} = 500$, no critical-region enhancement). Table 9 gives the result: phase equilibria is only $\approx 2.5\%$ of the stage time; the *cDFT interface solve dominates* ($\approx 97.5\%$). The per-point cost moreover broadly rises with the number of components — from 0.18 s/point for the 2-component feed to 1.8 s/point for the 8-component feed, a $\sim 10\times$ spread driven by the coupled multicomponent density profiles, not by the flash — though it is not strictly monotonic (the 4-component feed at 0.60 s/point exceeds the 6-component feed at 0.53 s/point), since per-point cost also depends on how near-critical and hard-to-converge a feed is. (These absolute times sit below the production-grid maps of Table 4 because the latter used the finer $n_{\text{grid}} = 2048$ grid; the $\approx 2.5\%$ phase-equilibria fraction and the component-count scaling are the grid-independent take-aways.)

Fe0s reference (Rehner et al. 2023). The Fe0s framework [1] — the same engine used to generate our γ maps — reports (1D, 1024 grid points, Intel Core i7-7700K) a single planar surface tension in 0.09 s and a 50-composition surface-tension diagram in 14.2 s (≈ 0.28 s/composition), with one Helmholtz-energy evaluation (with hyper-dual derivatives) at $\sim 0.9\text{--}1.5 \mu\text{s}$. The gap to

Table 6: 5-fold cross-validated RMSE for γ , in mN m^{-1} (mean $\pm$ std). TabPFN values are on the direct- γ target; GPR/SVGP on the residual target; SR on the reconstructed γ_{cDFT} scale. The scales differ (see text); bold marks the best full data-driven model.

Family	Model	RMSE _{CV} [mN m^{-1}]
Full data-driven	TabPFN without_HPO	0.1630 ± 0.0172
	TabPFN with_HPO	0.0395 ± 0.0094
Residual-correction	GPR (5000)	0.0379 ± 0.0262
	SVGP	0.0233 ± 0.0030
	SR	0.2475 ± 0.0110

Table 7: Time per state point for γ and speed-up vs. cDFT (mean cDFT cost 3.48 s/point). cDFT on c109 (AMD EPYC 9754); TabPFN on the NVIDIA H100; residual models on the local Intel Xeon Silver 4214R CPU. Cross-family numbers are therefore order-of-magnitude, not same-rig.

Model	Mode	Time/point	Speed-up vs. cDFT
cDFT (PC-SAFT)	per point (range)	0.75–6.57 s (mean 3.48 s)	$1\times$
TabPFN without_HPO	isolated	0.73 ms	$\sim 4\,700\times$
TabPFN with_HPO	isolated	1.74 ms	$\sim 2\,000\times$
TabPFN with_HPO	batched	0.48 ms	$\sim 7\,200\times$
GPR (RBF+White)	isolated	0.86 ms	$\sim 4\,000\times$
GPR (RBF+White)	batched	0.0086 ms	$\sim 406\,000\times$
SVGP (Matérn-5/2)	isolated	4.99 ms	$\sim 700\times$
SVGP (Matérn-5/2)	batched	0.047 ms	$\sim 74\,000\times$
SR (closed-form eq.)	batched	$\sim 8\times 10^{-6}$ ms	$\sim 4\times 10^8\times$

our 0.75–6.6 s/point is therefore *not* explained by the phase-equilibrium flash (which Table 9 shows is negligible, $\approx 2.5\%$): it is that our “point” is a coupled *multicomponent* (2–8 species) interface solve — intrinsically 10–70 $\times$ more expensive than the simple 1D pure-/binary-fluid interface that the reference times — on a finer grid. The order of magnitude is consistent once this is accounted for.

GPU-accelerated 3D cDFT (Stierle et al. 2024). Even the state-of-the-art GPU-accelerated cDFT of Stierle et al. [2] (JAX, backward-mode automatic differentiation, RTX 4090) requires *seconds* per calculation: one 3D equilibrium density profile takes 0.42 s ($N_{\text{grid}} = 8$) to 21.8 s ($N_{\text{grid}} = 256$), and a full 3D adsorption isotherm (20 pressures, $N_{\text{grid}} = 64$) takes ≈ 16 s. Their GPU implementation is ~ 50 – $60\times$ faster than a single CPU core and up to $550\times$ faster than the **Fe0s** CPU reference at $N_{\text{grid}} = 128$ (where **Fe0s** needs 1580 s). Table 10 places our numbers alongside these.

Take-away. GPU acceleration brings cDFT from minutes/hours to *seconds* per calculation [2]; our ML surrogates bring it a further 10^3 – $10^4\times$ down, to *sub-millisecond* per point — i.e. faster than even GPU-accelerated cDFT — while preserving cDFT-level accuracy ($R^2 \gtrsim 0.9999$). The two approaches are complementary: fast (GPU) cDFT generates the training data; the surrogate then serves predictions at a cost compatible with high-throughput screening and gradient-based inverse design.

Table 8: Resource comparison across the three approaches for γ . The full data-driven column lists **without_HP0**/**with_HP0** TabPFN (H100); the hybrid column is WSD + SVGP residual (1500 inducing points). cDFT has no training phase or stored model (“—”); **n/r** = not separately recorded. ^a cDFT per-point solve, default grid $n_{\text{grid}} = 500$ on **c109**, range over the 2–8-component feeds; inherently one solve per point (no batching). ^b isolated single-sample on the local Xeon Silver 4214R; batched deployment ≈ 0.025 – 0.047 ms/sample. ^c peak resident set size (Slurm **MaxRSS**) of the cDFT job. SVGP inference was measured isolated on the local Xeon. TabPFN runs on the H100, so its constraining memory is VRAM (~ 4.9 GB for **with_HP0**); cDFT and SVGP are CPU-only and use no VRAM. ^e SVGP training timing/RAM as recorded by the instrumented run on **c109** (96 cores): wall 880.2 s (≈ 14.7 min), aggregate CPU time 83 293 s across the cores, and 1724 MB peak resident set during training. ^d 5-fold CV mean (see Table 6 for $\pm \text{std}$); cDFT is the ground-truth reference (RMSE $\equiv 0$); SVGP RMSE is on the smaller-magnitude residual target, not directly comparable to the direct- γ TabPFN values.

Metric	PC-SAFT + cDFT	Full data-driven (TabPFN)	WSD + Residual (SVGP)
γ RMSE _{CV} [mN m ⁻¹]	0 (reference)	0.163 / 0.040	0.023 ^d
Training wall time [s]	—	1.31 / 1217.4	880.2
Training CPU time [s]	—	8.94 / 2790.7	83293 ^e
Per-sample inference [ms]	184–1791 ^a	0.734 / 1.735	5.23 ^b
Model size [MB]	—	87.9 / 50.4	9.1
Host RAM [MB]	622–908 ^c	27.6 / 3251.9	1724 ^e
GPU memory / VRAM [MB]	— (CPU)	86.1 / 4853.8	— (CPU)

Table 9: Re-timed cDFT γ map on **c109** ($2 \times$ AMD EPYC 9754; default grid $n_{\text{grid}} = 500$, hybrid 4×4 threading): phase-equilibria stage vs. cDFT interfacial sweep, per feed. “Total/pt” is the combined stage time per (T, P) point.

Feed	Comp.	Points	Phase-eq [s]	cDFT sweep [s]	cDFT/pt [s]	Total/pt [s]	Phase-eq share
0	2	95	1.6	15.8	0.167	0.184	9.2%
1	4	100	1.8	57.9	0.579	0.597	3.0%
2	6	100	2.2	51.1	0.511	0.533	4.1%
3	7	105	3.5	181.1	1.725	1.758	1.9%
4	8	100	3.1	176.1	1.761	1.791	1.7%
Total / aggregate		500	12.2	481.9	—	—	2.5%

8 Conclusions

- Across all four targets, both TabPFN variants reproduce cDFT with $R^2 \geq 0.9998$ on the test split; **with_HP0** is the more accurate and the right choice for deployment (CV γ RMSE = 0.040 mN m⁻¹).
- Cross-validation, not the i.i.d. test split, is the honest accuracy metric: the CV γ RMSE (0.040 mN m⁻¹) matches the deployment error on the full map (0.03 – 0.05 mN m⁻¹), while the test RMSE (0.006 mN m⁻¹) was $\sim 6\times$ optimistic.
- Replacing cDFT by the surrogate accelerates a γ map by 10^3 – $10^4\times$ (hours $\rightarrow$ seconds) at negligible accuracy cost.
- Residual-correction models also work well; on the residual target SVGP has the smallest CV RMSE (0.023 mN m⁻¹) and GPR/SR are the cheapest to serve ($\sim 8\mu\text{s}$ and $\sim 8\text{ns}$ per point batched).
- Direct re-timing on **c109** shows the cDFT per-point cost is dominated by the multicom-

Table 10: Literature context for the cost of a cDFT evaluation vs. our ML surrogate. Tasks differ (1D interface vs. 3D adsorption profile), so this is a qualitative placement, not a controlled comparison.

Source / method	Task	Wall time
This work, cDFT (FeOs , c109)	1D γ /point, 2–8 comp, $n_{\text{grid}}=2048$	0.75–6.6 s
This work, cDFT (FeOs , c109)	1D γ /point, 2–8 comp, $n_{\text{grid}}=500$ (re-timed)	0.18–1.8 s
Rehner 2023 [1], FeOs CPU	1D surface tension, 1024 pts	0.09 s
Rehner 2023 [1], FeOs CPU	50-composition σ diagram	14.2 s
Stierle 2024 [2], JAX GPU	3D density profile ($N=64$)	0.51 s
Stierle 2024 [2], JAX GPU	3D density profile ($N=256$)	21.8 s
Stierle 2024 [2], JAX GPU	3D isotherm (20 p , $N=64$)	≈ 16 s
Stierle 2024 [2], FeOs CPU	3D profile ($N=128$)	1580 s
This work, TabPFN/GPR/SR surrogate	1D γ per state point	5×10^{-4} – 1.7×10^{-3} s

ponent interface solve ($\approx 97.5\%$), not the phase-equilibrium flash ($\approx 2.5\%$); per-point cost scales with component count ($0.18 \rightarrow 1.8$ s for $2 \rightarrow 8$ species).

- Per state point, the learned models cost sub-millisecond vs. seconds for cDFT — faster than even the GPU-accelerated cDFT of the literature [2], while the physics engine (**FeOs** [1]) remains the source of ground truth.

References

- [1] P. Rehner, G. Bauer, and J. Gross, **FeOs: An Open-Source Framework for Equations of State and Classical Density Functional Theory**, *Ind. Eng. Chem. Res.* **62** (2023) 5347–5357. doi:10.1021/acs.iecr.2c04561.
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